No.
Date:

PR 6 MA1101 2025

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Jawaban

1. Turunan implisit $(y' = f(x,y))$

a. $(3xy+7)^2 = 6y$

$\Rightarrow 2(3xy+7)\left(\frac{d}{dx}3xy+7\right) = 6y'$

$\Rightarrow 2(3xy+7)(3(y+y'x)) = 6y'$

$\Rightarrow y' = (3xy+7)(y+y'x)$

$= 3xy^2 + 7y + 3x^2y'y + 7y'x$

$\Rightarrow y' - 3x^2y'y - 7y'x = 3xy^2 + 7y$

$\Rightarrow y'(1 - 3x^2y - 7x) = 3xy^2 + 7y$

$\Rightarrow \boxed{\frac{dy}{dx} = \frac{3xy^2+7y}{1-3x^2y-7x}}$

b. $x + \tan(xy) = 0$

$\Rightarrow 1 + (y+y'x)\sec^2(xy) = 0$

$\Rightarrow y\sec^2(xy) + xy'\sec^2(xy) = -1$

$\Rightarrow xy'\sec^2(xy) = -1 - y\sec^2(xy)$

$\Rightarrow \boxed{\frac{dy}{dx} = -\left(\frac{1+y\sec^2(xy)}{x\sec^2(xy)}\right) = -\frac{1}{x}\cos^2(xy) - \frac{y}{x}}$

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2. Turunan implisit dua kali $(y'' = f(x,y))$

a. $y^2 - 2x = 1 - 2y$

$\Rightarrow 2yy' - 2 = -2y'$

$\Rightarrow 2yy' + 2y' = 2 \Rightarrow yy' + y' = 1$

$\Rightarrow y'(1+y) = 1$

$\Rightarrow \dfrac{\partial y}{\partial x} = \dfrac{1}{1+y}$ (1)

Kemudian, cari turunan dari $\frac{\partial y}{\partial x}$ terhadap x

$\dfrac{\partial}{\partial x} \dfrac{\partial y}{\partial x}$

$\Rightarrow -\dfrac{y'}{(1+y)^2}$, Substitusi y' dengan (1)

$\Rightarrow \boxed{\dfrac{d^2 y}{\partial x^2} = -\dfrac{1}{(1+y)^3}}$

b. $2\sqrt{y} = x - y$

$\Rightarrow \dfrac{y'}{\sqrt{y}} = 1 - y' \Rightarrow y' = (1-y')\sqrt{y}$

$\Rightarrow y' = \sqrt{y} - y'\sqrt{y}$

$\Rightarrow y' + y'\sqrt{y} = \sqrt{y}$

$\Rightarrow y'(1+\sqrt{y}) = \sqrt{y}$ agar mempermudah kalkulasi

$\Rightarrow y' = \dfrac{\sqrt{y}}{1+\sqrt{y}} = 1 - \dfrac{1}{1+\sqrt{y}}$

$\Rightarrow \dfrac{\partial}{\partial x} y' = \dfrac{(\frac{\partial}{\partial x} 1 + \sqrt{y})}{(1+\sqrt{y})^2}$

$= \dfrac{\frac{1}{2\sqrt{y}} \cdot y'}{1+y+2\sqrt{y}} = \dfrac{\frac{\sqrt{y}}{1+\sqrt{y}}}{2\sqrt{y} \cdot (1+\sqrt{y})^2}$

$= \boxed{\dfrac{1}{2} \cdot \dfrac{1}{(1+\sqrt{y})^3}}$

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3. Pers $y = mx + b$ $(1,-1)$ dari

$(x^2+y^2)^2 = (x-y)^2 \Rightarrow (x^2+y^2)^2 = x^2+y^2-2yx$

$\Rightarrow 2(x^2+y^2)(2x+2yy') = 2x+2yy'-2(y+y'x)$

Malas mencari bentuk tertutup, substitusi:

$x=1,\ y=-1,\ y'=m$

$\Rightarrow 2\cdot 2(2+-2m) = 2-2m-2(-1+m)$

$\Rightarrow 8-8m = 2-2m+2-2m$

$\Rightarrow 4 = 4m$

$\Rightarrow m = 1$

$y = x+b \Rightarrow -1 = 1+b \Rightarrow b = -2$

Dengan demikian,

$\boxed{y = x-2}$

4. $xy + y^2 = 1$ di $(0,-1)$

$\Rightarrow y + xy' + 2yy' = 0$

$\Rightarrow xy' + 2yy' = -y$

$\Rightarrow y'(x+2y) = -y$

$\Rightarrow \frac{dy}{dx} = \frac{-y}{x+2y}$, nilai $y'(0,-1) = -\frac{1}{2}$

$\Rightarrow \frac{d^2y}{dx^2} = \frac{-(y'(x+2y) - y(1+2y'))}{(x+2y)^2}$

[- (min) di bawah itu dari faktor (-1), bukan hasil kuadrat]

$= \frac{(-\frac{1}{2})2(-1) - (-1)(1+2(-\frac{1}{2}))}{-4}$

$= \boxed{-\frac{1}{4}}$

$|1200 - 1600| = 4.00$

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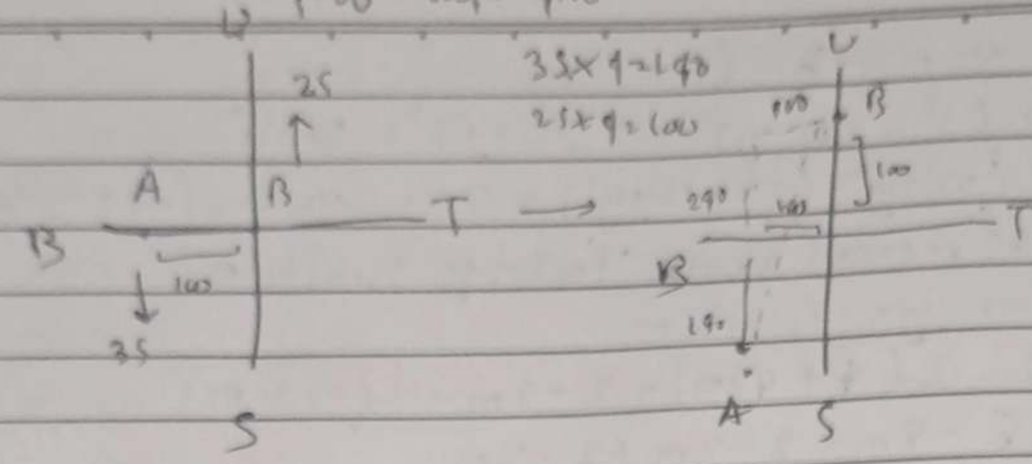

$|\vec{AB}|_{1200} = 100$ $\qquad$ $|\vec{AB}|_{1600} = 260$

Rumus jarak $\Rightarrow z^2 = x^2 + y^2$

Karena hanya ada perpindahan pada sumbu-y, misal total jarak $\Rightarrow y = a + b$ dengan $a = A$ (jarak), a' kecepatan
$b = B$ (jarak), b' kecepatan

$z^2 = x^2 + (a+b)^2$ terhadap waktu (t) dalam jam

$\Rightarrow 2zz' = 2xx' + 2(a+b)(a'+b')$

$\Rightarrow z' = \dfrac{(a+b)(a'+b')}{z}$

$= \dfrac{240\ (60)}{260}$

$= \dfrac{720}{13}$

$\approx \boxed{55{,}384615 \text{ km/jam}}$

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6. $y = 2\sin(\frac{\pi x}{2})$

Pakai rumus jarak dengan $x = \frac{1}{3}, y : 1$

$z^2 = x^2 + y^2 \quad \Rightarrow z = \sqrt{(\frac{1}{3})^2 + 1^2} = \frac{1}{3}\sqrt{10}$

$= x^2 + 4\sin^2(\frac{\pi x}{2})$

$\Rightarrow 2zz' = 2xx' + \frac{\pi}{2} 4.2\sin(\frac{\pi x}{2}).\cos(\frac{\pi x}{2}).x'$

$\Rightarrow z' = \frac{xx' + 4\pi \sin(\frac{\pi x}{2})\cos(\frac{\pi x}{2}).x'}{z}$

$= \frac{\frac{1}{3}.\sqrt{10} + 4\pi.\frac{1}{2}.\frac{1}{2}\sqrt{3}.\sqrt{10}}{\frac{1}{3}\sqrt{10}}$

$= \boxed{1 + \frac{3\pi\sqrt{3}}{2} \text{ cm/s}}$

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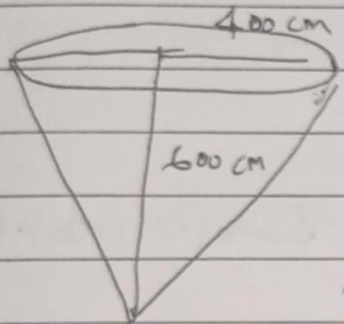
7.



$10.000 \text{ cm}^3/\text{m}$

$\Delta h = 20 \text{ cm/menit}$

$V' = V'_{masuk} - V'_{keluar} \quad (1)$

$V = \frac{1}{3}\pi r^2 h$

Pakai kesebandingan segitiga, didapat

$\frac{r}{h} = \frac{r_{max}}{h_{max}} = \frac{200}{600} = \frac{1}{3} \Rightarrow h = 3r \Rightarrow r = \frac{1}{3}h$

$\Rightarrow V = \frac{1}{3}\pi(\frac{1}{3}h)^2.h$

$= \frac{\pi h^3}{27}$

$\Rightarrow V' = \frac{3\pi h^2}{27}.h' = \frac{\pi h^2}{9}.20 = \frac{20\pi h^2}{9}$

Maka, laju $V'(200) = \frac{20\pi.h^2}{9}\Big]^{h=200}$

$= \frac{800000\pi}{9} \text{ cm}^3/\text{menit}$

Pakai (1), didapat $V'_{masuk} = V' + V'_{keluar}$

$= \boxed{10.000 + \frac{800000\pi}{9} \text{ cm}^3/\text{menit}}$

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8 $V = \frac{4}{3}\pi r^3$

$L = 4\pi r^2$, $\frac{dL}{dt} = 1$ cm^2/menit, $L = \pi d^2$

$\frac{dL}{dt} = 8\pi r . r'$ (?) $\frac{dL}{dt} = 2\pi d d'$

$\Rightarrow r' = \frac{1}{8\pi r}$ adalah laju turun ini $\Rightarrow d' = \frac{1}{2\pi d}$

$d = 10$ cm $\Rightarrow r = 5$ $d = 10$ cm

$\Rightarrow r'(5) = \frac{1}{8\pi(5)} . \frac{dd}{dr}$ $\Rightarrow d'(10) = \frac{1}{20\pi}$

$= \frac{1}{40\pi} . 2$ $\hookrightarrow d = 2r$

$dd = 2\,dr$

$= \frac{1}{20\pi}$ $\Rightarrow \frac{dd}{dr} = 2$

Dengan demikian, laju berkurangnya diameter adalah

$\boxed{\frac{1}{20\pi} \text{ cm/menit}}$